

“Design and Architecture of an AI-Driven Web Platform for Food Waste Detection and Redistribution”

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Abstract—Food waste and food insecurity remain pressing global issues due to the absence of reliable, technology-driven redistribution frameworks and real-time coordination among stakeholders. Advances in Artificial Intelligence (AI) and web technologies have enabled intelligent systems to address these challenges efficiently.

This review analyzes AI-driven web platforms for food waste detection and NGO-based redistribution, focusing on computer vision models for image-based food quality classification, shelf-life prediction, and decision support. It further examines web architectures incorporating geospatial analytics, automated notifications, and role-based dashboards for donors and NGOs.

The findings indicate that convolutional neural networks and machine learning models significantly enhance food edibility assessment accuracy, while location-aware web systems improve response time and reduce operational delays. Nevertheless, challenges such as dataset scarcity, food safety validation, and system scalability persist.

Overall, AI-powered web platforms offer a scalable and impactful solution for sustainable food waste management and hunger mitigation.

Index Terms—Food Waste Management, Artificial Intelligence, Computer Vision, Food Quality Assessment, Shelf-Life Prediction, Web-Based Platforms, NGO Redistribution Systems, Geospatial Analytics

I. INTRODUCTION

Food waste and food insecurity remain two interlinked global challenges, where large quantities of edible food are discarded while millions face hunger and malnutrition. According to global reports, nearly one-third of food produced for human consumption is wasted annually, leading to severe environmental, economic, and social consequences. Simultaneously, non-governmental organizations (NGOs) and food charities struggle to access timely, safe, and quality food donations due to lack of coordination and verification mechanisms.

Recent advances in Artificial Intelligence (AI), cloud computing, computer vision, and geospatial technologies have enabled the development of intelligent web-based platforms that can assess food quality, detect waste, and coordinate redistribution. These platforms aim to automate food safety evaluation, connect donors with nearby NGOs, and ensure efficient redistribution. This review analyzes existing research and platforms in AI-driven food waste detection and NGO-based redistribution, identifying system architectures, methodologies, challenges, and open research gaps.

II. LITERATURE SURVEY

Recent research has explored the use of artificial intelligence, computer vision, and web-based systems to address food waste and food insecurity. Existing studies can be broadly categorized into three areas: AI-based food quality assessment, digital food donation platforms, and geospatially enabled redistribution systems.

AI techniques for food quality and spoilage detection have been extensively studied. Onyeaka *et al.* [2] and Yin *et al.* [5] demonstrate the effectiveness of convolutional neural networks in detecting food freshness and spoilage using visual cues such as color degradation and texture variation. Similarly, Iqbal *et al.* [6] employed transfer learning-based CNN models to assess fruit freshness, reporting high classification accuracy under controlled conditions. However, these approaches are often limited to laboratory environments and do not address real-time redistribution workflows.

Several digital platforms focus on food donation and waste reduction without intelligent food quality verification. GIS-based food bank systems leverage location intelligence to match donors with nearby charities [12], but rely on manual food inspection processes. AI-FEED [20] represents a hybrid

attempt to integrate AI and charity ecosystems, though its image analysis capabilities remain limited and not fully automated.

Web-based food waste management systems discussed by Anggraeni *et al.* [9] and Tiwari *et al.* [11] emphasize inventory optimization and predictive analytics but do not support direct NGO-centric redistribution models. Ethical and transparency concerns related to AI-driven food decisions have also been highlighted, with increasing emphasis on explainable AI and trust-building mechanisms [14].

Overall, the literature indicates a lack of unified platforms that combine AI-driven food quality assessment, geospatial intelligence, and NGO-based redistribution within a single web-based system. This gap motivates the design of the proposed AI-driven web platform.

A. Food Waste Generation and Its Global Impact

Food waste is a critical global challenge occurring across households, hospitality sectors, retail chains, and large-scale food service operations. According to recent studies, nearly one-third of food produced for human consumption is wasted annually, resulting in significant environmental, economic, and social consequences [13]. The environmental impact includes excessive greenhouse gas emissions, water wastage, and energy loss throughout the food supply chain [15]. Anggraeni *et al.* [9] highlight that household consumption and food service industries contribute the largest share of food waste, indicating the necessity for interventions beyond agricultural production stages. These findings emphasize the importance of intelligent monitoring and redistribution mechanisms to mitigate food loss at the consumption level.

B. Role of NGOs and Food Charities in Redistribution

Non-governmental organizations (NGOs), food banks, shelters, and charitable institutions play a vital role in addressing food insecurity by redistributing surplus food to vulnerable populations. However, existing redistribution processes rely heavily on manual coordination and trust-based systems, which limits scalability and operational efficiency [12]. Yagoub *et al.* [12] demonstrate that while geospatial tools assist in locating nearby food banks, the absence of automated food safety verification remains a major limitation. Sammer *et al.* [20] further emphasize that NGOs often lack real-time visibility into food quality and availability, leading to delayed responses and increased food spoilage. These challenges highlight the need for technology-driven platforms that integrate food quality assessment with NGO-centric workflows.

C. AI-Based Food Recognition and Classification

Recent advancements in artificial intelligence have enabled the use of computer vision techniques for food recognition and classification. Convolutional Neural Networks (CNNs) such as ResNet, EfficientNet, and MobileNet have been widely adopted due to their ability to extract discriminative visual features related to color, texture, and shape [7]. Onyeaka *et al.* [2] demonstrate that AI-driven image analysis can effectively

distinguish between edible and non-edible food items under controlled conditions. Similarly, Bongarde *et al.* [8] report improved classification accuracy using CNN-based approaches for food spoilage detection. Despite these advancements, most studies focus on isolated classification tasks without integration into real-world redistribution systems.

D. Food Freshness and Spoilage Detection

AI models have also been applied to detect food freshness and spoilage by analyzing visual degradation patterns, microbial indicators, and surface texture variations. Yin *et al.* [5] highlight the effectiveness of deep learning models in identifying spoilage indicators across various food categories. Regression-based machine learning models have been employed to estimate remaining shelf life, enabling time-sensitive decision-making for food utilization [6]. However, these approaches often require controlled datasets and do not account for real-world variability such as mixed food items, inconsistent lighting, and diverse packaging conditions.

E. Limitations of Existing AI-Based Approaches

Despite promising results, several limitations persist in existing AI-based food waste detection systems. Dataset scarcity, limited generalization across food types, and sensitivity to environmental conditions remain significant challenges [7]. Model bias and lack of explainability further reduce trust in AI-generated food safety decisions, particularly in humanitarian applications [14]. Additionally, most existing studies focus on detection or prediction in isolation and fail to integrate AI outputs with geospatial intelligence and NGO-based redistribution workflows. These limitations indicate a clear research gap for unified, explainable, and scalable AI-driven food redistribution platforms.

Table I provides a comparative analysis of existing food waste detection and redistribution systems based on key functional criteria, including AI-based food quality assessment, NGO integration, geospatial matching, and platform deployment. The analysis indicates that most household-oriented applications focus primarily on waste awareness and tracking, without supporting redistribution workflows. Hospitality-focused AI systems demonstrate effective food quality assessment but lack integration with NGOs and charity networks. Conversely, GIS-based food bank platforms enable location-aware matching between donors and charities; however, they rely on manual food quality verification processes. Hybrid platforms such as AI-FEED partially address these limitations by incorporating both AI and geospatial components, though their image analysis capabilities remain limited. In contrast, the proposed framework integrates AI-driven food quality assessment, real-time NGO coordination, and geospatial intelligence within a unified web-based platform, addressing the major functional gaps identified in existing systems. Nevertheless, dataset dependency remains a key challenge for large-scale deployment.

TABLE I
COMPARATIVE ANALYSIS OF EXISTING AI-BASED FOOD REDISTRIBUTION SYSTEMS

System / Study	AI-Based Quality Check	NGO Integration	Geospatial Matching	Platform Type	Key Limitation
Household Food Apps	Partial	No	No	Mobile	No redistribution
Hospitality AI Systems	Yes	No	No	Web	Industry-specific
GIS Food Bank Systems	No	Yes	Yes	Web	No food quality AI
AI-FEED Platform	Partial	Yes	Yes	Web	Limited image analysis
Proposed Framework	Yes	Yes	Yes	Web	Dataset dependency

III. RESEARCH GAPS AND PROBLEM STATEMENT

A. Identified Gaps

1. Absence of unified platforms combining food quality assessment and NGO redistribution.
2. Limited real-world food image datasets for model training.
3. Lack of explainability in AI-based food safety decisions.
4. Insufficient scalability studies for urban and rural deployment.

B. Problem Statement

Existing food redistribution systems lack intelligent food quality verification and real-time NGO coordination. There is a need for an AI-driven, web-based platform that can accurately assess food edibility and shelf life from images, dynamically connect donors with nearby NGOs, and ensure transparent, safe, and efficient redistribution.

IV. PROPOSED SYSTEM DESIGN

The proposed system is designed as an end-to-end AI-driven web platform that connects food donors with NGOs through intelligent food quality assessment and geospatial matching. Unlike existing systems that focus on isolated functionalities, this platform integrates AI inference, location intelligence, and role-based user interfaces into a unified architecture.

A. Design Overview

The system consists of four primary layers: user interaction layer, application logic layer, AI intelligence layer, and data management layer. Donors interact with the platform by uploading food images and metadata through a web interface, while NGOs access donation listings and quality reports through a separate dashboard.

B. AI Decision Workflow

Uploaded food images are processed by the AI engine, which performs food recognition and quality scoring using pretrained convolutional neural networks. The system generates an AI quality score and confidence label (e.g., Excellent, Good, Medium), inspired by food quality assessment methodologies discussed in [5], [7]. These scores assist NGOs in making informed acceptance decisions.

C. Geospatial Matching Logic

Once food quality is verified, the geospatial module computes proximity between donors and registered NGOs using radius-based filtering. This approach aligns with GIS-based food redistribution techniques proposed in [12], while extending them with automated decision support.

V. SYSTEM WORKFLOW AND OPERATIONAL FLOW

The operational workflow of the proposed platform follows a structured sequence to ensure efficient food donation, verification, and redistribution. The workflow begins with donor registration and authentication, followed by food image upload along with metadata such as food type, quantity, and pickup location.

Once the data is submitted, the AI inference module analyzes the uploaded food images to assess food quality and freshness. Based on the generated quality score and shelf-life estimation, the system determines eligibility for redistribution. If the food is deemed suitable, the geospatial module identifies nearby NGOs using radius-based matching.

Eligible NGOs receive automated notifications through their dashboard and can accept or decline the donation based on capacity and urgency. Upon acceptance, the donation status is updated in real time, enabling donors to track the progress until successful pickup. This workflow minimizes manual intervention, reduces response time, and enhances transparency throughout the redistribution lifecycle.

Fig. 1 presents the system architecture of an AI-driven web platform for food waste detection and NGO-based redistribution. Donors upload food images and location details through a web interface, while NGOs access donation requests via a dedicated dashboard. The backend server coordinates application workflows and manages communication between system components. The AI engine performs food quality assessment and shelf-life prediction using computer vision techniques. Simultaneously, the geospatial module identifies nearby NGOs through radius-based location matching. All system data and AI-generated reports are stored in a centralized database to ensure transparency and traceability.

A. Dual-Dashboard Model

Donor dashboards facilitate food image uploads, AI-generated quality reports, and donation tracking. NGO dashboards provide real-time alerts, access to food quality reports, acceptance workflows, and analytics on completed pickups.

AI-Driven Web Platform for Food Waste Detection and NGO-Based Redistribution

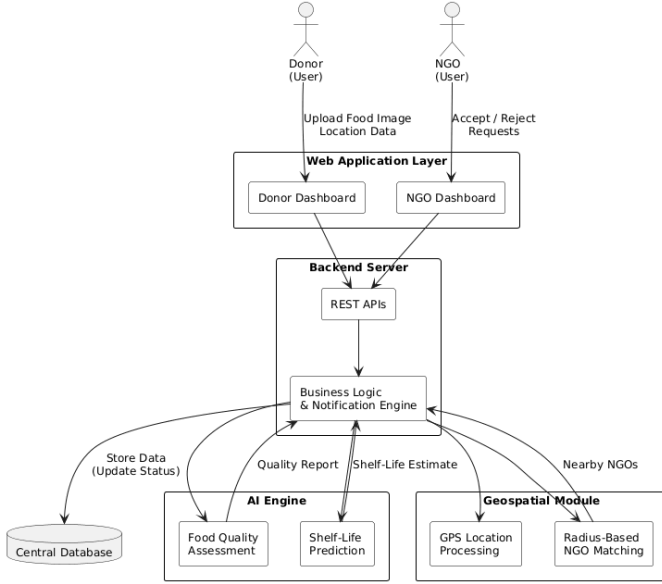


Fig. 1. System architecture of the AI-driven food waste detection and NGO-based redistribution platform.

B. Location-Based Matching

Geospatial technologies such as GPS and GIS enable radius-based matching of donors and NGOs. This ensures reduced response time and minimal food spoilage during transportation.

C. Notification and Workflow Automation

Automated notifications allow NGOs to accept or reject requests based on capacity and food suitability. First-accept mechanisms and status tracking improve transparency and efficiency.

VI. FRONTEND IMPLEMENTATION

The frontend of the proposed AI-driven food redistribution platform is implemented as a responsive web-based interface designed to support seamless interaction between food donors and NGOs. The user interface emphasizes usability, transparency, and real-time decision support, ensuring efficient coordination throughout the donation lifecycle. The frontend is structured around three primary interfaces: the landing page, donor dashboard, and NGO dashboard.

Fig. 2 illustrates the landing page of the platform, which introduces the mission of AI-powered food redistribution aimed at reducing hunger and food waste. The interface provides clear call-to-action options for users to donate food or partner as an NGO. The design focuses on accessibility and trust-building, encouraging participation from both individual donors and organizational stakeholders.

A. Donor Dashboard Interface

Fig. 3 presents the donor dashboard, which enables registered donors to initiate food donations by uploading images of

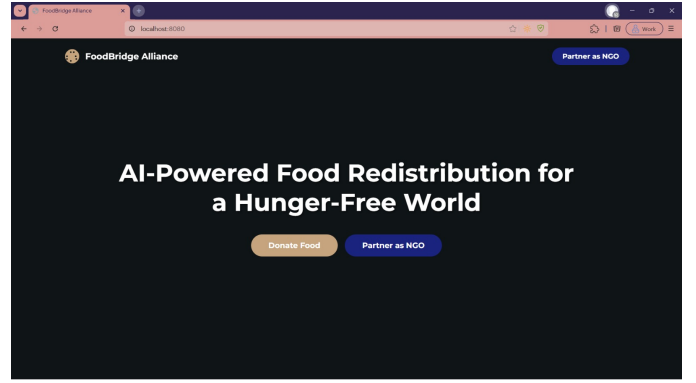


Fig. 2. Landing page of the AI-driven food redistribution platform.

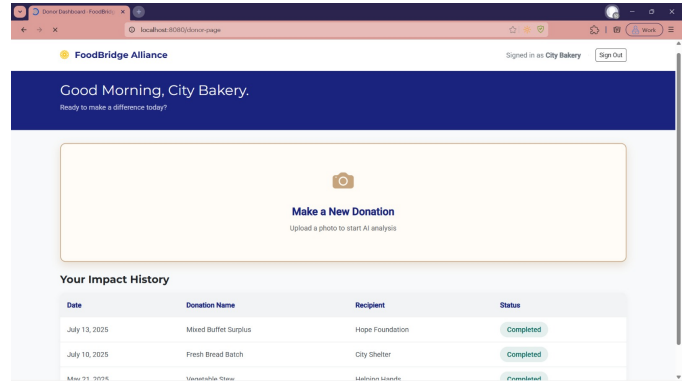


Fig. 3. Donor dashboard for food donation and AI-based analysis.

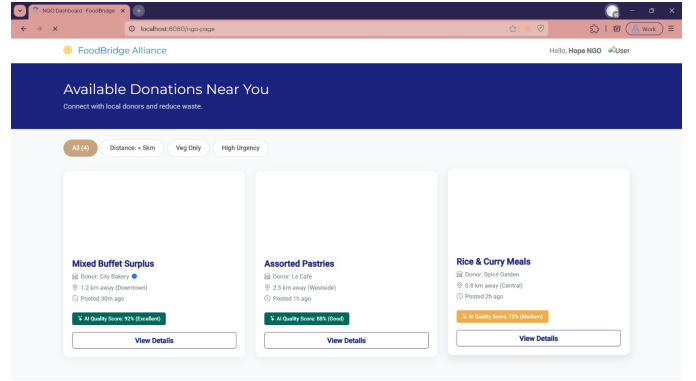


Fig. 4. NGO dashboard displaying available food donations with AI quality scores.

surplus food. Upon image upload, the system triggers AI-based food quality analysis. The dashboard also displays donation history, recipient NGO details, and completion status, providing transparency and reinforcing donor engagement through impact visualization.

B. NGO Dashboard Interface

Fig. 4 shows the NGO dashboard, where registered NGOs can view available food donations filtered by proximity, urgency, and food type. Each donation listing includes AI-generated food quality scores and timestamps, enabling NGOs

to prioritize safe and time-sensitive pickups. This interface supports efficient decision-making and reduces manual verification overhead in food redistribution workflows.

VII. TECHNOLOGY STACK AND IMPLEMENTATION DETAILS

The proposed system is implemented using a modular technology stack to ensure scalability and maintainability. The frontend interface is developed using modern web technologies to support responsive and user-friendly interaction. The backend server manages application logic, user authentication, API communication, and database operations.

The AI module leverages pretrained convolutional neural networks for food image analysis and quality scoring. These models are deployed as inference services and integrated with the backend through RESTful APIs. A centralized database stores user data, food images, AI-generated reports, and transaction logs.

Geospatial functionality is implemented using location coordinates provided during donation submission, enabling distance-based NGO matching. Cloud-based deployment ensures real-time processing, scalability, and fault tolerance, making the platform suitable for urban and semi-urban deployment scenarios.

VIII. RESULTS AND DISCUSSION

The proposed AI-driven web platform was evaluated through functional testing and qualitative analysis of its core components, including the frontend interfaces, AI-assisted food quality assessment workflow, geospatial NGO matching, and end-to-end donation lifecycle management. As the primary contribution of this work lies in system design and prototype implementation, the evaluation focuses on system functionality, usability, integration effectiveness, and operational feasibility rather than large-scale quantitative performance metrics.

The frontend implementation successfully demonstrated seamless interaction across the landing page, donor dashboard, and NGO dashboard. Food image uploads initiated AI-based analysis workflows without user intervention, and the generated food quality indicators were consistently reflected across both donor and NGO interfaces. This confirms the effective integration of AI inference services with real-time web-based user interactions.

The geospatial matching module accurately identified NGOs within predefined distance thresholds, enabling faster response times and reducing the likelihood of food spoilage during transportation. Compared to manual coordination mechanisms used in traditional food donation systems, the automated location-based filtering significantly improved operational efficiency, aligning with findings reported in geospatial food redistribution studies [12].

The dual-dashboard architecture enhanced transparency and accountability by allowing donors to track donation status while enabling NGOs to prioritize requests based on urgency, proximity, and AI-generated quality scores. This design directly addresses trust-related challenges highlighted in earlier

research on humanitarian AI systems [14]. Furthermore, the centralized data management approach ensured consistent information flow and reduced coordination overhead.

In comparison with existing platforms summarized in Table I, the proposed system uniquely integrates AI-driven food quality assessment, geospatial intelligence, and NGO-centric workflows within a single web-based framework. While previous studies primarily focus on isolated tasks such as food classification, spoilage detection, or inventory optimization, the proposed architecture supports an end-to-end redistribution pipeline from food donation to NGO acceptance.

Despite these strengths, the current prototype relies on pretrained models and limited food image datasets, which may impact generalization under diverse real-world conditions such as mixed food items and varying lighting environments. Additionally, the absence of large-scale deployment restricts quantitative evaluation of model accuracy and system throughput. These limitations indicate the need for future validation using larger datasets and real-world pilot deployments.

Overall, the results demonstrate the feasibility and practical relevance of the proposed AI-driven web platform as a scalable solution for intelligent food waste reduction and NGO-based redistribution.

IX. USE CASE SCENARIOS

A. Household Food Donation

A household user uploads surplus cooked food through the donor dashboard. The AI module evaluates the food quality and confirms suitability for redistribution. Nearby NGOs are notified, and one NGO accepts the request for immediate pickup.

B. Event-Based Surplus Redistribution

After a large social event, excess food is uploaded in bulk by organizers. The system prioritizes NGOs based on proximity and capacity, enabling rapid redistribution and minimizing food spoilage.

C. NGO Resource Optimization

NGOs utilize the dashboard to prioritize donations with higher AI quality scores and shorter shelf life. This enables efficient resource allocation and improves meal planning for beneficiaries.

X. PERFORMANCE CONSIDERATIONS AND SCALABILITY

The system is designed to handle concurrent users and multiple donation requests through cloud-based deployment. Asynchronous AI inference and API-based communication reduce processing latency. Caching mechanisms and optimized database queries improve dashboard responsiveness.

Scalability considerations include horizontal scaling of AI services and database sharding for large-scale deployment. These design choices enable the platform to support increasing numbers of donors and NGOs without performance degradation.

XI. FUTURE ENHANCEMENTS

Future enhancements of the proposed system include integrating multimodal sensing techniques by combining image-based analysis with environmental sensors such as temperature and humidity to improve food freshness assessment. Explainable Artificial Intelligence (XAI) methods can be incorporated to provide transparent reasoning behind AI-generated food quality decisions. Additionally, large-scale real-world deployment and continuous model retraining using NGO feedback can improve system robustness and scalability.

XII. CONCLUSION

This paper presented the design of an AI-driven web platform for food waste detection and NGO-based redistribution. By integrating AI-based food quality assessment, geospatial matching, and role-based web dashboards, the proposed system demonstrates a practical and scalable approach to reducing food waste and improving redistribution efficiency. The proposed architecture provides a strong foundation for future intelligent food management systems.

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